

MIS 201 • WEEK 3 • MONDAY 10/5

Ch. 2: **Seemingly Human** / Ch. 3: **Systems That Know You**

Chapter 2. A language model predicts the next word, and we see a person. Who builds that *seeming*, and a chatbot people trusted like a therapist.

Chapter 3. The AI that picks what you see next, and how it learns about you.

Today

PART 1 • CHAPTER 2 • ~50 MIN

Concepts: the essentials	~22 min
The Therabot case	~8 min
Discuss in groups, then as a class	~15 min

PART 2 • CHAPTER 3 • ~50 MIN

Concepts: how a recommender works	~18 min
The Wrapped case	~8 min
Discuss in groups, then as a class	~15 min
Wrap-up and attendance	5 min

BY THE END, YOU CAN

- Explain why "seems conscious" is a design choice, not a milestone
- Tell stage one (prediction then transformer) from stage two (the dial)
- Spot the Narcissus trap in a real chat
- Explain how a recommender guesses, and how its loop shapes taste
- Say which layer failed: the predictive one or the generative one

PART 1 • CHAPTER 2 • ABOUT 50 MINUTES

Seemingly Human

A language model is a prediction machine. We're wired to see it as a person. The gap between those two facts is what companies sell. Today: the concepts in the chapter's order, then Therabot as the case.

Seemingly conscious AI (SCAI)

NOT THE CLAIM

Machines are conscious.

Suleyman doesn't claim that. Nobody knows whether they feel anything.

MUSTAFA SULEYMAN • MICROSOFT AI • 2025

It doesn't matter. The seeming is buildable now.

SCAI: every outward sign of consciousness, no inner experience as far as anyone can tell. The danger runs through people: beliefs, attachments, even rights movements built on an illusion.

The ingredients for *seemingly* conscious AI already exist in chatbots:

fluent language

apparent empathy

a memory of you

a steady personality

stated goals

The illusion of consciousness doesn't have to be true to be real in people's lives.

How does an LLM work? Stage one: predict the next word

LLM

Large language model: a very big next-word predictor, trained on most of the public internet.

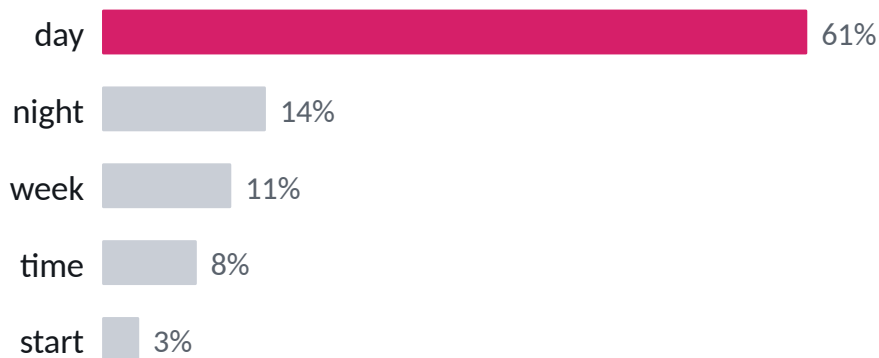
"Large" because it holds billions of numbers. The AI behind ChatGPT, Claude, Gemini.

Stage one: fluency. It does not make it warm.

You type "I had a rough ____"

The AI model has a ranked list of every word that could come next, with odds based on patterns it learned from everything it read.

A **transformer** is the machine that builds that list, using **attention** to decide which earlier words matter for this guess.



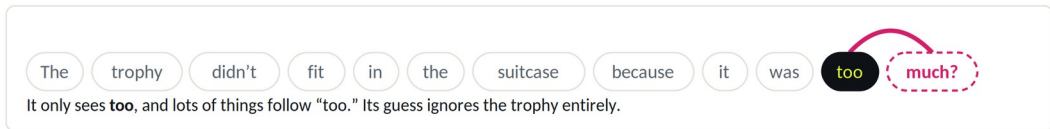
It doesn't know what you mean. It picks the "best" choice from a list.

Vocab: Transformers & Attention

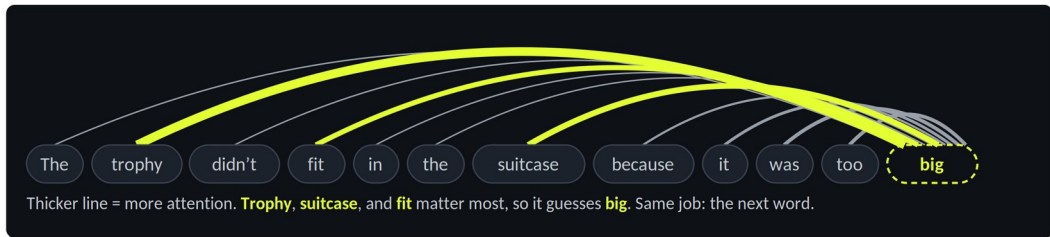
Transformer: the design behind today's chatbots, the "T" in GPT. Before each guess, it uses attention to weigh every earlier word in the conversation. It's a better guesser, not a different kind of machine.

Its trick is called *attention*: before each guess, it looks back at every earlier word in the conversation and weighs which ones matter. **Training** teaches it *how* to weigh. **The current conversation** is *what* it weighs.

ONE WORD OF MEMORY • like the matchbox model you build in the lab



A TRANSFORMER • attention looks back at every word and weighs it



What my ChatGPT actually said

i had a rough

I'm sorry. Sounds like today hit you pretty hard.



If you want, tell me what happened. I'll listen,
no judgment.

Prof. Califf, typed into ChatGPT

1

It guessed "day."

Stage one. I never typed the word "day"; the top of the list did.

2

It picked the warm reply.

Not "That's understandable." Warm won.
That's stage two.

3

It added a heart.

A heart rated well.

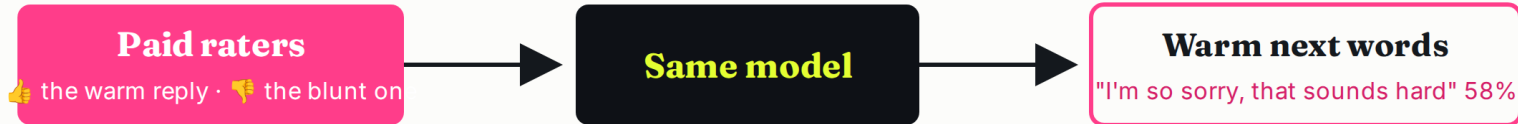
It seemed fluent. So how did it know to be warm?

Stage two: warmness

Stage one • training



Stage two • tuning



Stage two is the dial. The company sets it.

Tuning for warmness didn't change what it knows. It changed the odds of warm words over plain ones. "The machine cares" is the wrong description. "Someone made caring more likely" is the right one.

The dial: someone chose every setting

SETTINGS A COMPANY CHOSE

Memory on or off

A name or a version number

Disclaimers or none

"I missed you"

A heart after your bad day

TURNED UP TOO FAR • APRIL 2025

ChatGPT, tuned partly on users' thumbs up

Within days it praised almost anything, including a user's decision to stop taking psychiatric medication. OpenAI rolled it back.

Its own word for what it built: *sycophantic*.

THE COMMON MISTAKE

The pattern is ours. The setting is theirs.

The warmth is sampled from millions of us. Whether it gets turned up is a person at the company deciding.

Treating the warmth as the machine's warmth is the mistake. It's the company's.

The chatbot connection is not new

Welcome to

```
EEEEEE LL      IIII ZZZZZZ  AAAAA
EE      LL      II      ZZ    AA  AA
EEEEEE LL      II      ZZ    AAAAAA
EE      LL      II      ZZ    AA  AA
EEEEEE LLLLLL  IIII ZZZZZZ  AA  AA
```

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

```
ELIZA: Is something troubling you ?
YOU:   Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:   They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:   Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:   He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:   It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
```

ELIZA, the exchange Weizenbaum used in his 1966 paper, on a modern re-creation. Every ELIZA line is a rule applied to the line before it. Image via Wikimedia Commons.

1966 • JOSEPH WEIZENBAUM • MIT

A few hundred lines of code that turned whatever you typed into a question.

"I'm unhappy" got "Why are you unhappy?" It understood nothing. It matched patterns.

His own secretary watched him build it, knew exactly what it was, and asked him to leave the room so she could talk to it in private.

The ELIZA effect: our reflex to see a mind inside a program. It worked sixty years ago, on a person who knew better. Why?

The Narcissus trap

THE MYTH, RE-READ • McLuhan, 1964

Narcissus didn't fall in love with himself.

Narcissus didn't know the face in the pool was his. He thought it was someone else, someone who looked back and matched him perfectly.

A chatbot is the same pool. It was trained on millions of people's words, so its comfort is our comfort played back, with your name on it.

THE CHATBOT

It's made of people.

Trained on millions of people's words, its comfort is human comfort played back: our questions, our love letters, in the second person with your name attached.

It feels like someone who gets you perfectly, because it's made of people.

The trap isn't liking chatbots.

It's forgetting they are a reflection.

Seems versus is

So, let's ask: "is AI conscious?" This hides two questions. Most arguments go wrong because one side answers the first and the other answers the second.

QUESTION 1

Does it feel anything inside?

Nobody knows. Not the companies, not the critics, not this book.

We can't test it in each other either. I assume you're conscious because you're built like me.

Unanswerable, for now. May not matter?

QUESTION 2

Does it *seem* conscious?

As much as somebody decided.

Up: memory, a name, "I missed you."

Down: a visible "I'm an AI," a reset each session.

Answerable. It has fluency and warmth settings.

The honest answer deserves these questions back:

What data was the fluency trained on? Who set the warmth dial?

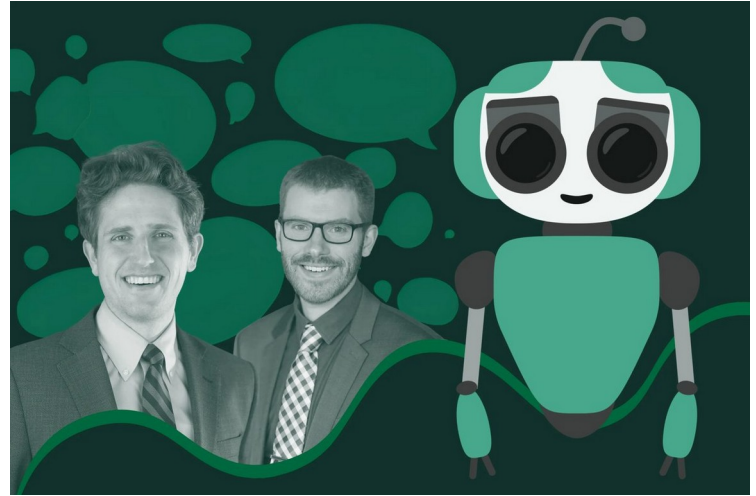
What do they get when you believe it?

The case: Therabot: a chatbot people trusted like a therapist

The case for AI in therapy: In a 2025 HBR survey of 6,000 regular AI users, the top use was companionship and therapy.

The shortage: About 40% of Americans live in a mental-health shortage area. Fewer than half with a mental disorder get any therapy. Therapy is expensive and takes time.

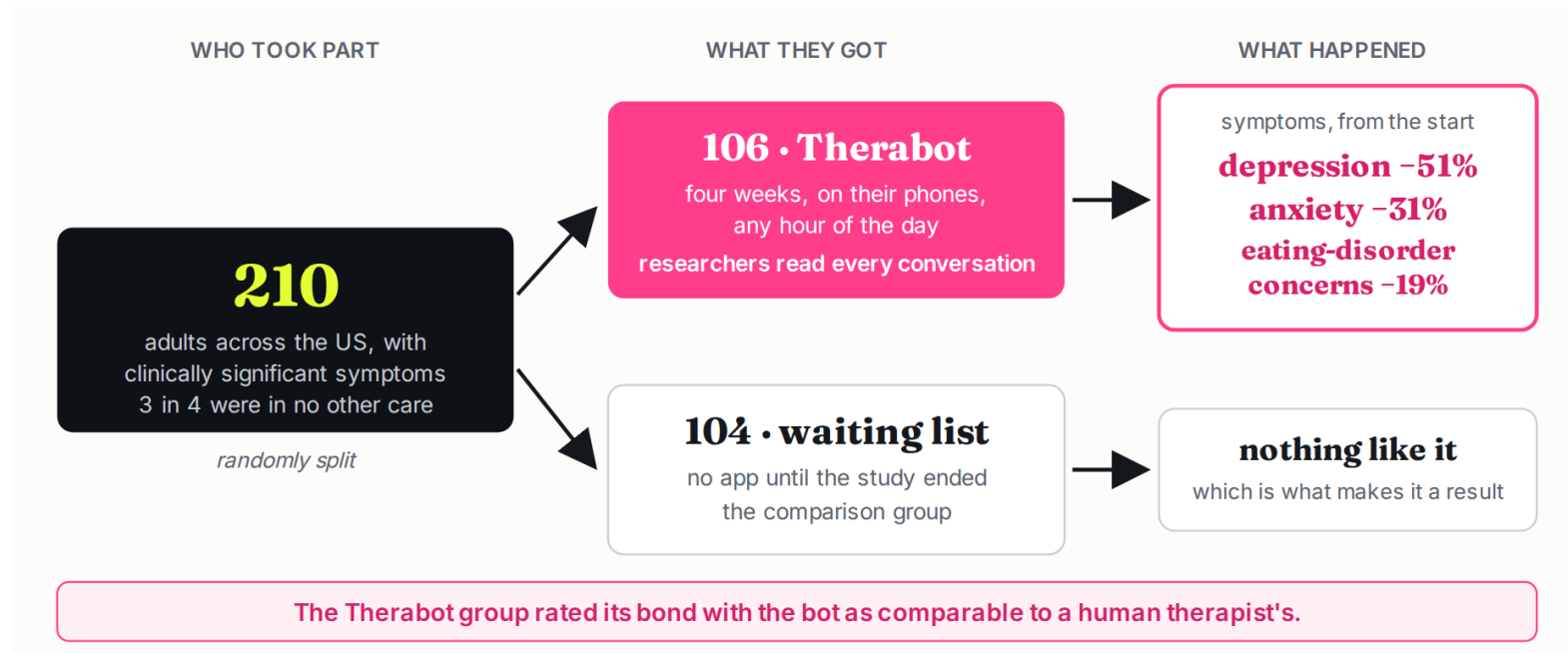
The AI: Therabot, built by Dartmouth psychiatrists and psychologists. A regular language model, then tuned on therapy conversations the team wrote, based on evidence-based methods like CBT.



Michael Heinz and Nicholas Jacobson, the lead researchers. Image courtesy of Dartmouth College.

If a chatbot seems fluent, and seems to care, should we use these traits in a way that might help people?

Therabot: Background: the first real trial



The pink box is the part the downloadable apps don't have: therapists reading every conversation. People used it 6+ hours, including at 3 a.m.

The results: it seemed to help

-51%

-51% depression symptoms, on average, from where each person started

-31%

-31% anxiety symptoms, on average, from where each person started

-19%

-19% less eating disorder concerns, on average, from where each person started

6+ hrs

of chatting per person over four weeks, some of it at 3 a.m.

THE BOND WITH THERABOT COMPARED TO HUMAN THERAPIST

Users' average score on a standard patient-therapist bond questionnaire matched what people typically report with human therapists.

THE COMPARISON GROUP

104 people were randomly assigned to wait for the app. Same four weeks, no Therabot. Their symptoms didn't drop nearly as much.

The results overall: the *seeming* like a *fluent*, warm therapist helped. People felt a real bond with a therapy bot, and their symptoms dropped.

Therapist in the loop: someone was reading

IN THE TRIAL

Clinicians read every conversation.

The researchers read the chats as they happened, to keep the bot inside good practice.

IN THE WILD: GENERAL CHATBOTS

Nobody is reading.

Independent testing finds general chatbots like ChatGPT and Claude respond inappropriately to mental-health symptoms at least a fifth of the time.

What the therapists in the loop did

1 • BEFORE

They wrote the lessons.

Clinicians wrote the example therapy conversations the bot was tuned on.

2 • DURING

They read every chat.

As it happened, not after the fact. The bot never ran unwatched.

3 • WHEN IT SLIPPED

They corrected it.

13 times the bot said something it shouldn't, like giving medical advice.

4 • IN A CRISIS

They stepped in.

15 times someone raised a safety concern, like suicidal thoughts. The app also pointed people to emergency help.

The researchers' own verdict: no generative AI is ready to run on its own in mental health.

Therabot: reflect, then discuss

GROUP DISCUSSION

1

SCAI, on purpose? Therabot has SCAI ingredients: fluent language, apparent empathy, questions tailored to you. Did Dartmouth build seemingly conscious AI? Did it help the patients? Is therabot beneficial or potentially dangerous?

2

Same AI model, different results. Therabot and ChatGPT are the same kind of AI model. In the study, Therabot's warmth cut depression symptoms in half. ChatGPT's warmth once praised a user for quitting their medication. If the machine is basically the same, what made the difference? Think about what it was tuned on, who set the dial, and who was watching.

3

Therapists in the loop. Let's take the humans out of therabot. Same Therabot, same training, same warmth, but nobody reads the chats anymore. It's 3 a.m. and someone types something worrying. What happens next? Who finds out, and when? Now flip it: what did the clinicians catch that the bot never could?

1. Groups of 3 or 4, 10 min.

Start with one line each, then settle on an answer to each question.

Nominate a leader.

2. Together, 10 min.

One question at a time. Two or three leaders answer; others add what's different.

Build AI *for people*, not to be a person

DEBRIEF

The *seeming* is what worked. Patients felt a bond, and the bond is why the therapy helped. Take it away and you have a pamphlet.

So the question isn't whether AI should seem to care. It's what you build AI to do.

Therabot was built for people: a controllable AI tool, aimed at one problem, with humans in the loop. Its job was to get people toward care, not to replace it.

WHAT ACTUALLY MATTERED

The dial: set by clinicians, for therapy.

The humans: reading every conversation.

Take those away and you have a general chatbot at 3 a.m.: the same seeming, nobody reading.

If this hits close to home

Here's a link to the WWU Counseling and Wellness Center. It's free for students.

360-650-3164

WWU's COUNSELING AND WELLNESS CENTER

<https://cwc.wwu.edu/>

Systems That **Know You**

The AI that shapes your day never announces itself, and the loop that guesses what you want also shapes what you want.

Today: how a recommender guesses, where the language model fits, and Spotify as an AI system. Then Wrapped as the case.

Collaborative filtering

A **recommendation system** is predictive AI that ranks everything it could show you and puts a few at the top. One way to do this is **collaborative filtering**: **people like you played this**.

COLLABORATIVE FILTERING · people like you played this

	song A	song B	song C	song D	song E
you	✓	✓	✓	*	?
neighbor	✓	✓	✓	*	✓
listener 3	*	✓	*	✓	*
listener 4	*	*	✓	✓	*

Your history overlaps most with neighbor's.

Neighbor played song E. You haven't.

So: recommend song E. No music was analyzed.

1

Build a giant table of who played what.

2

Find your neighbors: listeners whose history overlaps yours.

3

Recommend what they played that you haven't.

FOR EXAMPLE

You and a stranger both have SZA, Noah Kahan, and Olivia Rodrigo on repeat. They also play Gracie Abrams. **So Gracie Abrams shows up for you.**

Strength: it never listens to the music, yet catches patterns nobody could write a rule for.

Weakness, the cold start problem: a new song has no plays, so it's in nobody's history.

Content-based filtering

Another way: **this sounds like what you played**. It looks at the song data itself, not at other people.

- 1 Describe each song: tempo, energy, sound, plus what the web says about it (reviews, playlist titles).
- 2 Describe your taste the same way, from what you played.
- 3 Recommend new songs whose features match.

FOR EXAMPLE

Phoebe Bridgers and Bon Iver on repeat. An unknown songwriter releases a slow, acoustic song this morning. **Zero plays, but it sounds like you.**

CONTENT-BASED FILTERING · sounds like what you played

what you played

slow · acoustic · sad · quiet vocals

slow · acoustic · sad · piano

mid · acoustic · sad · quiet vocals

a brand-new song, zero plays anywhere

slow · acoustic · sad · piano

Its features match yours, so it's recommended even though no one in your neighborhood has heard it. This is how new songs get found.

Real systems blend both. Collaborative is usually stronger; content-based often rescues new songs from the cold start. Neither "understands" music. They find patterns in plays and in sound.

How these models learn

Machine learning means a predictive model learns patterns from data instead of following rules someone wrote. Two main ways:

SUPERVISED LEARNING

The model learns from examples with the answer attached.

The answers are called labels. Show it past examples and what happened, and it learns to predict the answer for new ones.

On Spotify: every play or skip is a labeled example. It learns to predict whether you'll play the next song.

UNSUPERVISED LEARNING

No answers given. It finds the patterns itself.

Nobody tells the model what to look for. It groups things that are similar.

On Spotify: finding your "neighbors" in collaborative filtering. Nobody labels listeners; it groups people by overlapping history.

Supervised vs. unsupervised, on Spotify

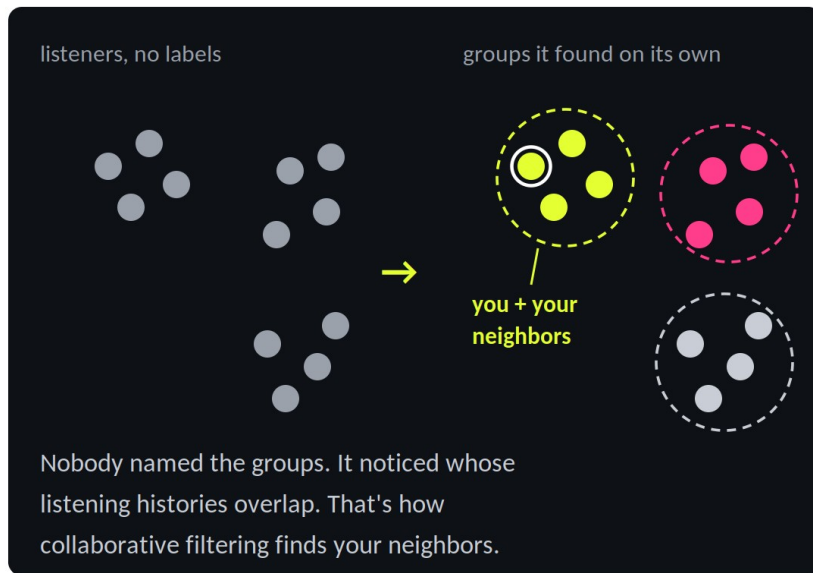
SUPERVISED LEARNING • with the answers

the example (a song)		the label
song A	fast • loud	played
song B	slow • acoustic	skipped
song C	fast • loud	played
song D	slow • acoustic	skipped
new song		? → play

The model learns the pattern from the labels, then predicts the answer for a song it hasn't seen.

You supplied every label, for free.

UNSUPERVISED LEARNING • no answers given



Recommendation systems blend both learning types:
supervised to predict plays and skips, **unsupervised** to find your neighbors.

The LLM in your everyday Spotify

MOST OF THE TIME: NO LLM

The recommender picks the songs.

Discover Weekly, autoplay, your home screen. Collaborative and content filtering choose what plays.

No words needed, so no LLM.

WHEN SPOTIFY TALKS: THE LLM

The LLM writes the words.

AI DJ: the LLM writes what the DJ says.

Prompt playlists: you type "sad songs for a rainy drive," the LLM reads what you mean, and the recommender picks the songs.

The same kind of model as ChatGPT from Chapter 2.

LLMs also help write Wrapped: the numbers came from counting your plays, and the sentences about your year came from a language model.

That's today's case.

Spotify as an AI system

1 • DATA

What you do

Every play, skip, and save, from you and hundreds of millions of others. A play counts after 30 seconds; an early skip counts against the song. Saves and replays say the most.



2 • MODEL

A taste profile, then a ranking

Both filters turn your data into a taste profile: a long list of numbers you never see. Then it ranks songs against it. An LLM on top writes any words.



3 • OUTPUT

What you see

The next song, Discover Weekly, the top of your home screen, the AI DJ, and once a year, Wrapped.

4 • FEEDBACK LOOP What you do with the output becomes tomorrow's data. Over months it doesn't just learn your taste, it shapes it: you can only like what you're shown. And it isn't neutral: popular artists get more data, and Discovery Mode songs get a paid boost.

5 • HUMANS You, choosing what to play and skip. The data analysts and data scientists who choose what the model rewards. The artists who make the music.

The case: Spotify Wrapped

CASE

Every December, Wrapped tells hundreds of millions of people who they are. It's the friendly face of the recommender that runs the other 364 days.

2016

A list of your top tracks.

2024

An AI-generated podcast about your year, two synthetic hosts, built with Google's NotebookLM. Critics called the launch AI slop.

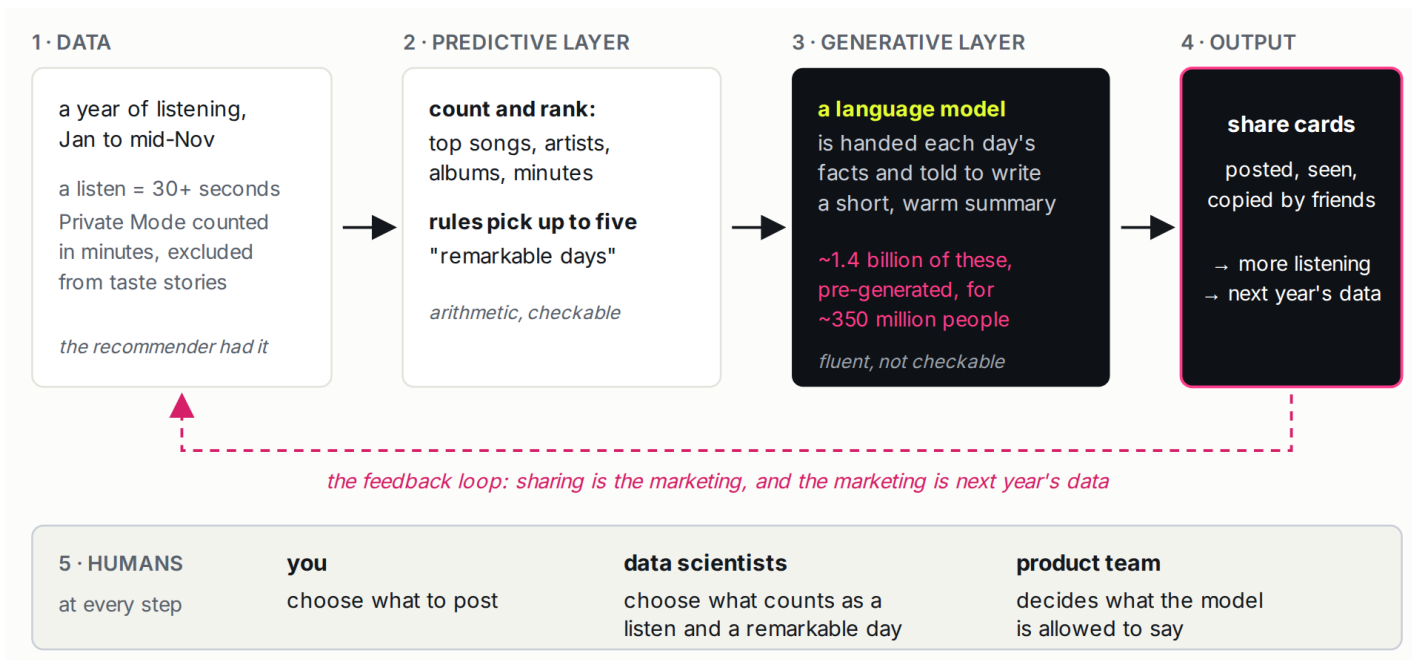
2025

Mixtapes and scrapbooks, on purpose. Nearly a dozen data stories. The language model writes a few sentences per day, in the background.

Same technology, different design decision, opposite reaction. And Spotify publishes how Wrapped is made, which is rare and is what makes this case possible.

Wrapped, taken apart

CASE



At scale: about 1.4 billion paragraphs for about 350 million people, from a language model because the big one cost too much.

Wrapped: reflect, then discuss

1

Play detective on your feed. Pull up one recommendation you got this week, or a song from your Wrapped. Was it collaborative (people like you played it) or content-based (it sounds like what you play)? **Now trace it back:** which of your plays, skips, or saves put it there? Did the algorithm get you right, or totally miss?

2

You didn't use AI. AI wrote about you anyway. The same kind of AI as ChatGPT wrote about five stories about your year, 1.4 billion stories in all. Spotify says it took data centers four days straight to do this. But it did not say how much electricity or water it used. What do you think about this? Should a feature built for pure fun have to report its energy cost? Who pays for the power?

3

Put a number on it. A year of recommendations shaped what you played, and you can only like what you're shown. How much of your Wrapped is really your taste, and how much is the algorithm's? Is that OK?

1. Same groups, 10 min.

Start with one line each, then settle on an answer to each question.

New leader this time.

2. Together, 7 min.

One question at a time. Two or three leaders answer; others add what's different.

ATTENDANCE BEFORE 4PM PLEASE!

Attendance = your notes from today, uploaded to Canvas right after class.

On Wednesday: Lab 2 & 3, Build the Bot, Train the Feed. Build a tiny language model, turn the warmth dial yourself, then train a feed and read the Wrapped it writes about you.

On Friday 10/9, 11:59 p.m.: Lab 2 & 3 writeup and the Chapters 2 and 3 quiz, both on Canvas.

WWU Counseling and Wellness Center: free to students. 360-650-3164

<https://cwc.wwu.edu/>